Journal of Quantitative Spectroscopy & Radiative Transfer 353 (2026) 109841

Contents lists available at ScienceDirect

Journal of Quantitative Spectroscopy and Radiative Transfer

journal homepage: www.elsevier.com/locate/jqsrt

Continuum absorption of millimeter waves by CH_4-X (X=CH_4, N_2, CO_2)

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ARTICLE INFO

Keywords:
Green-house gases
Methane
Collision-induced absorption
Bimolecular absorption
Resonator spectroscopy
Millimeter waves

ABSTRACT

The room-temperature continuum absorption of pure methane and of its mixture with nitrogen and carbon dioxide was measured in the millimeter wavelength range using resonator spectrometer. The line wings contribution to the continuum was estimated to be weak. The bimolecular nature of the observed continuum was confirmed by both the expected pressure dependences and the agreement with trajectory-based simulation.

1. Introduction

Methane is a potent greenhouse gas that plays a critical role in Earth's radiative balance and atmospheric chemistry [1]. Beyond Earth, CH_4 is an important atmospheric constituent on Titan [2] and within the gas giants such as Jupiter [3], where it significantly influences atmospheric chemistry, energy transport, and climate dynamics. In exoplanetary science, methane absorption features serve as key diagnostics for determining atmospheric composition, thermal structure, and the presence of disequilibrium chemical processes [4].

As a spherical top molecule with tetrahedral symmetry, methane lacks permanent electric dipole moment. Nevertheless, it exhibits a complex resonance spectrum arising from structural deformations due to anisotropic centrifugal distortion and from specific degenerate vibrations ([5] and references therein). It also shows broad collision induced absorption (CIA) resulting from the transient dipole moment induced during molecular interactions – a phenomenon that is known to contribute significantly to opacity in dense planetary atmospheres [6].

Accurate atmospheric studies therefore demand high-precision spectroscopic calculations across broad temperature ranges. Seminal CIA calculations for CH_4 were performed by Borysow and Frommhold [7] employing a method based on effective isotropic potentials combined with empirically adjusted short-range induced dipoles. These semiempirical results, extended later to CH_4-N_2 mixture [8], remain widely used. While Borysow's semiempirical data demonstrate good agreement with modern, *ab initio* classical trajectory-based calculations [9] for temperatures above 100 K, a pronounced discrepancy emerges at lower temperatures. This divergence has a tangible impact, substantially affecting the modeling of Titan's far infrared radiance. To date, no prior study has calculated the CIA for CH_4-CO_2 molecular pairs employing *ab initio* potential energy and induced dipole surfaces. To address this, the accompanying study [10], presents a new 5D induced dipole surface based on coupled-cluster level quantum chemistry computations and provides first-principle, trajectory-based results across a broad temperature range. This new dataset for CH_4-CO_2 is poised to become an important input for radiative transfer models for an accurate characterization of the greenhouse effect in CO_2-rich atmospheres containing methane, such as the early Martian paleoatmosphere or similar exoplanetary environments.

Experimental validation of these theoretical results remains critical. Although measurements are available in the far-infrared for both CH_4-CH_4 and CH_4-N_2 systems, data in microwave spectral range remain scarce. To our knowledge, the sole millimeter-wave continuum study of pure CH_4 and CH_4 mixture with nitrogen was performed at room temperature in the 30–750 cm^{-1} range by Birnbaum et al. [11]. In [12] measurements of the microwave absorption of CH_4-N_2 CIA have been reported at 4.85 cm^{-1} between 126 and 212 K, and in [13] were conducted measurements of CH_4-CO_2 CIA at 2.3 cm^{-1} at room temperature. Concerning temperature range coverage, the range between approximately 130 and 300 K has been relatively well studied for both CH_4-N_2

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https://doi.org/10.1016/j.jqsrt.2026.109841
Received 19 December 2025; Received in revised form 21 January 2026; Accepted 22 January 2026
Available online 24 January 2026
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and CH_4-CH_4 CIAs (see [11,12,14] and [15,16], respectively). For CH_4-CO_2, measurements exist at room temperature [17,18] and across the 200–300 K range [19], although the latter suffered from considerably larger uncertainty. Further investigations at lower temperatures are encouraged, and measurements at higher temperatures may also be of interest for specific exoplanetary scenarios.

This paper is organized as follows. Experimental details are given in Section 2. Data analysis and continuum absorption retrieving are discussed in Section 3. The details of trajectory-based calculations are given in Section 4. Section 5 is devoted to uncertainty analysis, comparison with calculation results and continuum nature discussion.

2. Experimental details

Spectra of pure CH_4 and its mixture with N_2 and CO_2 were recorded with the resonator spectrometer [20] following the approach applied in our previous studies of bimolecular absorption of nonpolar molecules [21–24]. The absorption coefficient is traditionally found by the variation of the resonance curve width of a high-quality Fabry-Perot resonator after its filling with the studied gas. Three backward wave oscillators (BWOs) of the OB-71 (70–120 GHz), OB-24 (156–257 GHz) and OB-30 (247–354 GHz) type were used as radiation sources. The BWO frequency was stabilized by a Phase Locked Loop (PLL) against a harmonic of a microwave synthesizer (Anritsu, MG3692C). The resonance curve of a chosen eigen-mode was recorded by fast digital scanning of the BWO frequency around the corresponding eigen-frequency without phase jumps using a direct digital synthesizer PTS ×10 supplied as a reference oscillator for PLL [25]. A low-barrier planar Schottky diode was used as a detector at frequencies below 260 GHz whereas QMC InSb bolometer cooled by liquid helium was employed at higher frequencies. Kr and Ar served as non-absorbing gases for resonator intrinsic loss measurements, since their refractive indexes are close to those of CH_4 and CH_4-CO_2 (Kr) and CH_4-N_2 (Ar) mixture. This allowed minimizing the pressure difference between the studied and non-absorbing gas recordings keeping the same eigen-mode frequency without the change in the resonator length. The purity of all gas samples declared by the manufacturer was better than 99.99%. Gas pressure was monitored with two capacitance monometers: MKS Baratron Type 625B and Pfeiffer CCR 361 with full-scale ranges of 2000 and 1000 Torr, respectively, combined with MKS PR4000B readout. The declared uncertainty was 0.25 and 0.2% of the Reading, respectively. Seven standard Pt-100 sensors were installed inside the resonator for continuous monitoring of the gas and resonator mirrors temperature which was 296.5(3) K during all performed recordings. The experimental conditions are presented in Table 1. The extremely weak pure CH_4 absorption could be measured only at the highest possible pressure of 1500 Torr (limited by the capabilities of the gas chamber). CH_4-N_2 and CH_4-CO_2 absorption was measured at different mixture contents in order to check the continuum pressure dependence.

For each series of recordings from Table 1, several spectra were registered under permanent conditions and at different positions of the waveguide line (including radiation source and detector) relative to the resonator. The final spectrum (Fig. 1) was obtained by averaging these multiple recordings for minimizing the influence of the standing waves on the measured absorption value ("etaloning" effect) [20].

Table 1
Experimental conditions.

Series number	Frequency range, GHz	Perturber	P_{CH_4}, Torr	$P_{perturber}$, Torr	Number of spectra
1	70–120	CH_4	1500.9	-	4
2		N_2	601	626	
3		CO_2	1012.9	399.5	2
4	156–257	CH_4	1501.0	-	16
5		N_2	300.5	1075	4
5			451.1	850.1	
7			600.8	626.3	
8		CO_2	698.3	710	
9			496.3	978.7	
10			989.8	454.2	
11	247–354	CO_2	704.7	709.9	

3. General approach to continuum spectra analysis

The observed absorption α_{total} is the sum of the contributions of the monomer resonance lines, α_{res}, and continuum, α_{cont}: $\alpha_{total} = \alpha_{res} + \alpha_{cont}$. Several lines of the main isotopologue of methane $^{12}CH_4$ were detected in the studied frequency range despite their low intensity and significant pressure broadening (Fig. 1). The intensity of the strongest one is 2.657×10^{-28} cm/molecule [26]. However, the far wings of the neighbouring high-lying transitions with intensities up to 2×10^{-26} cm/molecule could contribute to the measured absorption α_{total}. The absorption of CH_4 lines was calculated using the Van Vleck-Huber profile [27] and HITRAN2020 parameters [26]. For lack of information on N_2-broadening of mm-wave CH_4 lines, they were considered to be similar to the air-broadening which is confirmed, for example, by comparison of Pine's measurement results of CH_4-N_2 broadening coefficients [28] with the HITRAN2020 data for CH_4-air [26]. The broadening coefficients of CH_4 lines by CO_2 were obtained by multiplying by 1.3 the HITRAN air-broadening coefficients following analysis from [29]. The presence of water impurity in the studied gas sample was revealed by observing its rotational line near 183 GHz. Water amount was estimated as 2.5 mTorr using the integrated intensity from HITRAN.

The continuum absorption α_{cont} of methane was determined as a difference of measured and calculated resonance absorptions. It is known to be a result of the intermolecular interactions of real gas molecules. The "efficiency" of double and triple intermolecular interactions could be estimated by second and third virial coefficients $B(T)$ and $C(T)$ of the real gas state equation:

$$\frac{PV_m}{RT} = 1 + \frac{B(T)}{V_m} + \frac{C(T)}{V_m^2} + \ldots, \tag{1}$$

where P is the gas pressure at temperature T, R is the universal gas constant, and V_m is the molar volume. Since $B(T)/V_m$ is about 2 orders of magnitude larger than $C(T)/V_m^2$ for both pure CH_4 and its mixtures with N_2 and CO_2 at room temperature [30], our analysis can be limited by the pair interactions only. Note that the higher the pressure, the more monomers form paired states. The resulting deviation of monomer concentration from the ideal gas approach can be quantified as $\delta n = -2K_2n^2kT$, where K_2 is the dimerization equilibrium constant. For methane at room temperature and pressure of about 2 atm, this correction is about 1% of the monomer concentration. However, given the significant noise level in the spectra recordings (Fig. 1), we deliberately omitted pressure adjustments for dimerization effects during data analysis.

The continuum absorption produces due to the paired collisions between methane molecules with each other and with X molecules (X is CH_4, N_2 or CO_2) as well as X molecules with each other. It can be written as

$$\begin{aligned}\alpha_{cont}(P,f) &= \alpha_{CH_4-CH_4}(P,f) + \alpha_{CH_4-X}(P,f) + \alpha_{X-X}(P,f)\\ &= C_{CH_4-CH_4}(f)P_{CH_4}^2 + C_{CH_4-X}(f)P_{CH_4}P_X + C_{X-X}(f)P_X^2,\end{aligned} \tag{2}$$

where f is frequency, $C_{CH_4-CH_4}(f)$, $C_{CH_4-X}(f)$ and $C_{X-X}(f)$ are pressure-normalized continuum coefficients. The second and third terms of Eq. (2) vanished for pure CH_4 and the remaining part corresponds to methane self-continuum. Measurements of the nitrogen and carbon dioxide self-continuum coefficient value $C_{CO_2-CO_2}(f)$ and $C_{N_2-N_2}(f)$ are presented in [23,24] supported by classical trajectory calculations.

In the mm-wave spectral range, continuum absorption of a pair of interacting X-Y molecules can be approximated by the quadratic frequency law for practical use [31]:

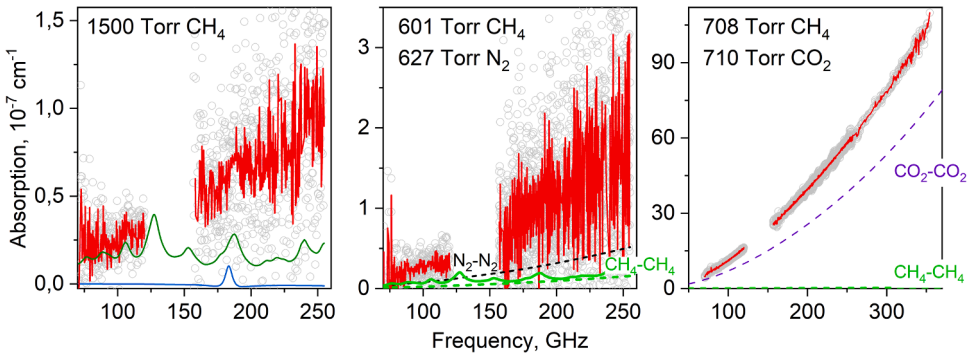

Fig. 1. Measured absorption spectra in methane (left panel) and its mixture with nitrogen (center panel) and with carbon dioxide (right panel) are shown in grey and their averages are in red. Calculated CH_4 and H_2O resonance absorption is demonstrated by green and blue curves. Bright green curves in the central and right panels correspond to pure CH_4 contribution to the measured absorption. Dotted curves are calculated contributions of CIA of CH_4, N_2 and CO_2.

$$C_{X-Y}(f) = C^0_{X-Y}\left(\frac{f}{c}\right)^2, \tag{3}$$

where C^0_{X-Y} is the continuum coefficient (or continuum cross section) and c is the speed of light. Thus, the continuum reduced spectral function defined in the mm-wave spectral range as $\alpha_{cont}(P,f)/f^2$ is considered constant. State-of-the-art precise measurements supported with theoretical results [23,24] and data analysis [32] demonstrated that N_2, CO_2 and H_2O continuum absorption does not follow this empirical representation due to bimolecular spectral features. The applicability of this approximation to the methane continuum is discussed in Section 4.

4. Trajectory-based simulation of CH_4-N_2 and CH_4-CO_2 CIA

The binary absorption coefficient for the pair X-Y is calculated using the following expression [33]

$$\alpha_{X-Y}(\nu) = m_{X-Y}\frac{(2\pi)^4 N_L^2}{3hc}\nu\left(1 - e^{-\frac{hc\nu}{k_B T}}\right)G(\nu), \tag{4}$$

where N_L is the Loschmidt constant, ν is the wavenumber, h and k_B are Planck and Boltzmann constants, respectively. Factor m_{X-Y} equals to $\frac{1}{2}$ when X and Y are identical molecules (the self-continuum), and 1 when X and Y are different species (the foreign continuum). Spectral density, $G(\nu)$, is obtained via the Fourier transform of the induced dipole time correlation function, $C_{corr}(t)$ [33]:

$$G(\nu) = \frac{1}{2\pi}\int C_{corr}(t)e^{2\pi i c\nu t}dt. \tag{5}$$

The classical line shape requires correction to satisfy the detailed balance condition prior to comparison with experimental profiles. For both the CH_4-N_2 and CH_4-CO_2 pairs, we applied Schofield's [34] and Frommhold's [33] desymmetrization procedures. It should be noted that in the millimeter-wave range, the correction to the spectral profile introduced by the desymmetrization procedure is minute.

While the complete computational methodology is detailed for CH_4-N_2 in [9] and for CH_4-CO_2 in [10], the essential details are summarized here. The spectral calculations for both molecular pairs employed potential energy surfaces computed at CCSD(T)-F12/aug-cc-pVTZ level of theory and induced dipole surfaces at the CCSD(T)/aug-cc-pVTZ level. The corresponding code implementing these surfaces for CH_4-N_2 and CH_4-CO_2 is available as C/C++ routines in Refs. [35] and [10], respectively. The correlation function in Eq. (2) was sampled at a fixed interval of 4.8 fs for CH_4-N_2 and 4.29 fs for CH_4-CO_2. A staged multi-temperature reweighting technique [36] was employed to ensure the convergence of trajectory-based profiles across the studied temperature range. Ensembles of up to 40 million trajectories were used for unbound states and up to 400,000 for bound states. Finally, the smoothing algorithm combining a weighted local regression over an adaptively sized window was applied to the resulting spectra [37].

5. Results and discussion

a. CH_4-CH_4

The methane monomer resonance lines contribute 10–60% to the total observed absorption in pure CH_4 (Fig. 1). The continuum of pure methane obtained after subtraction of the calculated resonance part from the measured absorption is seen from Fig. 2 to be well reproduced by the quadratic frequency dependence. Taking into account its

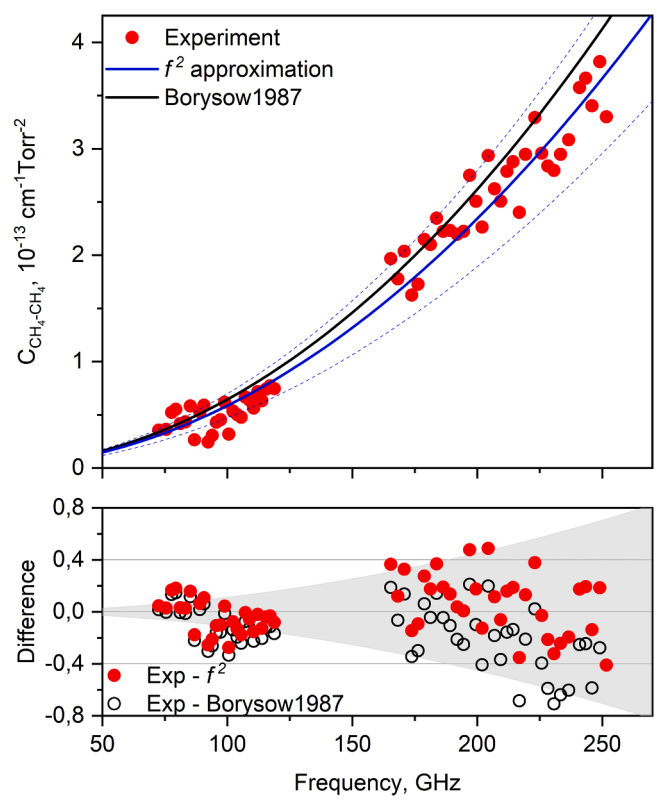

Fig. 2. *Upper panel*: Red points – CH_4-CH_4 continuum absorption. Blue solid line – f^2 approximation with corresponding uncertainties shown by dotted lines. Calculated absorption [7] is plotted by black. *Lower panel*: Difference (in 10^{-14} $cm^{-1}Torr^{-2}$ units) of measured continuum from f^2 approximation (red circles) and calculation [7] (open black circles). Grey area corresponds to f^2 approximation uncertainty.

smoothness, each point in Fig. 2 was obtained by averaging 6 neighbouring frequency points of the obtained continuum for improving the quality. The self-continuum coefficient $C^0_{CH_4-CH_4}$=5.3×10^{-16} cm/Torr^2 was retrieved from the fit of the quadratic function (3) to the experimental data. Its statistical uncertainty evaluated as 1 σ (σ is standard deviation) of experimental points from the function is 1×10^{-16} cm/Torr^2. The uncertainties associated with pressure instability during recordings and accuracy of our pressure gauges (see Table 1) were within 0.2%. The temperature variations were within 0.2 K during all the measurements; therefore, the related uncertainties can be neglected considering the temperature dependence from [7].

The uncertainty of the spectroscopic parameters of CH_4 lines significantly impacts the result. The line intensities and self-broadening coefficients in HITRAN have uncertainties more than 20% (according to HITRAN error codes). This leads to a 9–27% uncertainty in methane resonance absorption modeling depending on frequency. The corresponding $C^0_{CH_4-CH_4}$ uncertainty is 12.7%. It should be noted that HITRAN estimation of error codes could be rather conservative [38,39] and our estimate of the uncertainty due to line parameters corresponds to its upper limit.

The vagueness of the far-wing modeling makes it difficult to accurately estimate their contributions to the continuum. The variation of the wing cut-off from 25 to 4.8 cm^{-1} (which corresponds to the limits where impact approximation is still valid) alters the resonance spectrum by less than 17%, corresponding to 0.8 % variation in the continuum value.

All the uncertainty sources are summarized in Table 2. The total uncertainty of the obtained continuum coefficient was found as the square root of the sum of all aforementioned uncertainties squared (assuming their independence) and was equal to 23.1%. However, for accurate reproduction of the total absorption, it is usually recommended to use continuum data in conjunction with the parameters of the resonance spectrum with which this continuum was obtained. In this case, only experimental errors (statistical, temperature, pressure) will make an impact. Note also that the statistical uncertainties due to the experimental noise dominate the other error sources except the CH_4-CO_2 continuum case.

Our measurement results are in a good agreement with bimolecular absorption calculations by Borysow and Frommhold [7]. They differ from f^2 approximation of our $C^0_{CH_4-CH_4}$ by maximum 12% at the highest wavenumber, which is within the statistical uncertainty of our approximation (Fig. 2). This also confirms that the contribution of the far wings to the continuum is small and suggests that the continuum of pure CH_4 is determined mainly by bimolecular absorption.

b. CH_4-N_2

The contribution of the CH_4 lines broadened by N_2 pressure (1–26% of the total absorption, depending on CH_4 and N_2 partial pressure ratio and frequency) remains indistinguishable from the experimental noise. These lines were subtracted from the total absorption to retrieve the continuum. The methane self-continuum was calculated using our results from the previous section and contributed up to 8.5% to the observed absorption (Fig. 1). The nitrogen self-continuum [24] is responsible for 23–64% of absorption. The remaining part of the absorption (32–58%) is caused by the CH_4-N_2 interactions.

The CH_4-N_2 continuum absorption coefficient is presented in Fig. 3. Similarly to the self-continuum, each point in Fig. 3 is a result of averaging of 7 neighbouring frequency points of the obtained continuum for better quality. Quadratic frequency dependence works for this gas mixture as well. $C^0_{CH_4-N_2}$ obtained by the fit of Eq. (3) to the experimental data is 3.2×10^{-15} cm/Torr^2 with the statistical uncertainty of 7.7×10^{-16} cm/Torr^2. The uncertainties associated with the pressure changes (see Table 1) are within 0.5%. The temperature variations within 1 K during all the measurements and the related uncertainties are about 1.1% of the continuum coefficient value.

The impact of the high uncertainty of the calculated CH_4 monomer absorption is about twice less in the case of CH_4-N_2 continuum (~4%) than for the CH_4-CH_4 continuum due to a lower relative contribution of the resonance spectrum to the total absorption. The variation of the line wings cut off (in the same way as for the $C^0_{CH_4-CH_4}$ uncertainty estimation) changes $C^0_{CH_4-N_2}$ only by fractions of a percent. $C^0_{CH_4-CH_4}$ uncertainty (from Table 2) results in 1% uncertainty of $C^0_{CH_4-N_2}$. The accuracy of the N_2 absorption approximation [24] gives 4.2% CH_4-N_2 continuum uncertainty. The detailed uncertainty source analysis is summarized in Table 2. The total uncertainty of $C^0_{CH_4-N_2}$ is 47.2%.

The agreement of the CH_4-N_2 spectral function with the classical trajectories calculation [9] within 12% confirms the bimolecular nature of the observed continuum and serves as a validation of the calculation results. (Recall that [8] and [9] results give the same value of CH_4-N_2 CIA in the mm-wave spectral range).

Table 2
Uncertainty budget of the continuum coefficient.

Error source	Relative contribution in %		
	CH_4-CH_4	CH_4-N_2	CH_4-CO_2
Statistical	19.3	24.1	1.3
Pressure	0.2	0.5	0.3
Temperature	<0.1	1.1	<0.1
CH_4 resonance spectrum	12.7	4.0	<0.1
Far wings	0.8	0.4	<0.1
N_2 continuum	-	4.2	-
CO_2 continuum	-	-	1.4
CH_4-CH_4 continuum	-	0.7	<0.1
Total uncertainty	23.1	24.8	1.9

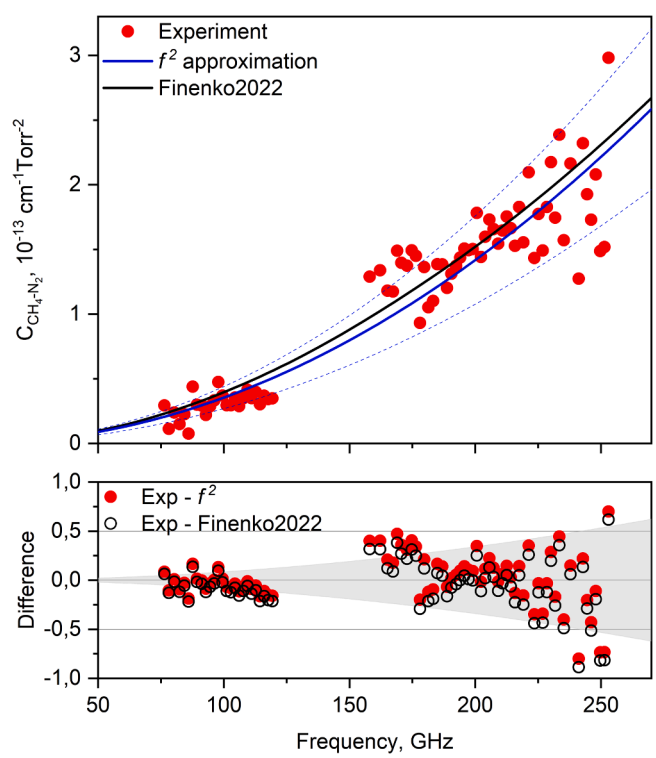

Fig. 3. *Upper panel*: Red circles – CH_4-N_2 continuum absorption with f^2 approximation shown by blue line with the uncertainty by blue dotted curves. Calculated absorption [9] is plotted by black solid curve. *Lower panel*: Difference (in 10^{-13} cm^{-1}Torr^{-2} units) of measured continuum from f^2 approximation (red circles) and calculation [9] (black open circles). Grey area corresponds to f^2 approximation uncertainty.

c. CH_4-CO_2

The absorption of carbon dioxide in the frequency range under consideration [23] is almost two orders of magnitude higher than that of methane for the pressures considered in this work. The contribution of the CH_4 resonance spectrum to the observed absorption is no more than 0.3%. The methane self-continuum contributed up to 1.6% to the observed absorption (at maximum CH_4 partial pressure). The carbon dioxide self-continuum was calculated using the approximations of our previous measurements [23]. This gives 41–84% of the observed absorption depending on CO_2 partial pressure. The remaining part of the absorption (16–58%) is caused by the CH_4-CO_2 interactions. The resulting CH_4-CO_2 continuum coefficient is presented in Fig. 4. Note that it is about one order of magnitude stronger than $C^0_{CH_4-N_2}$ due to the larger polarizability of the CO_2 molecule compared to N_2 [40].

In contrast to CH_4-CH_4 and CH_4-N_2 cases, the frequency dependence of $C^0_{CH_4-CO_2}$ deviates markedly from the quadratic law. Our measurements, however, are in excellent agreement with the recent trajectory-based calculation [10], exhibiting a mean difference of 1.4% with a standard deviation of 3.9%. As established in Ref. [10], the deviation is attributed to the CH_4-CO_2 dimer R-branch. This attribution stems from detailed trajectory-based calculations in which the contribution from bound states was isolated, revealing that the resulting frequency dependence within this region is predominantly due to resonant transitions within the dimer.

The uncertainties associated with the pressure changes (see Table 1) are within 0.3%. The temperature was very stable during CH_4-CO_2 continuum recordings (variations within fractions of a kelvin) and related uncertainties of the continuum coefficient value can be neglected. The small contribution of the CH_4 resonance spectrum to observed absorption leads to negligible errors of $C^0_{CH_4-CO_2}$ due to the inaccuracy of line modeling. The accuracy of the CO_2 absorption model [23] is a main contributor to the CH_4-CO_2 continuum error budget that gives 1.4% uncertainty. The detailed uncertainty source analysis is summarized in Table 2. The total uncertainty of $C^0_{CH_4-CO_2}$ is 1.9%.

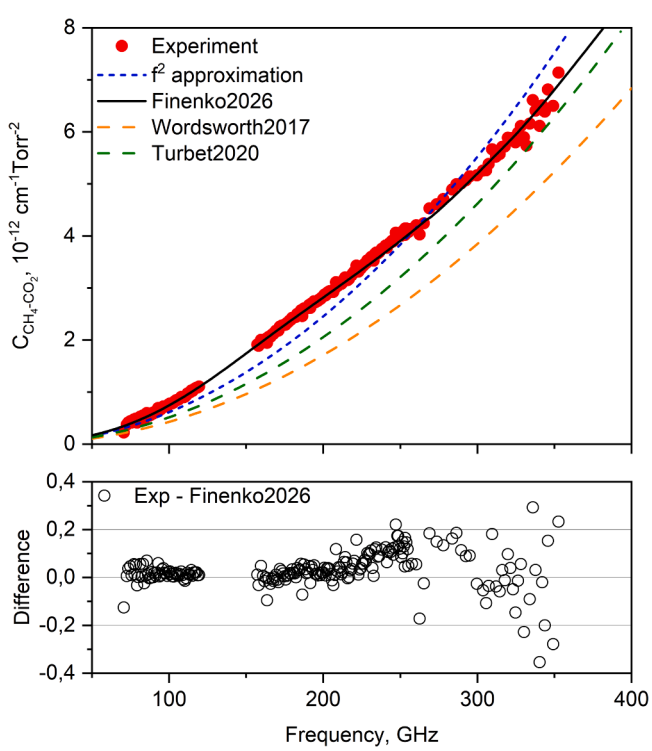

Fig. 4. *Upper panel*: Red circles – CH_4-CO_2 continuum absorption with f^2 approximation shown by blue dotted curve. Black solid curve and orange and green dotted curves are calculated absorption by [10,40,18]. *Lower panel*: Difference (in 10^{-12} cm^{-1}Torr^{-2} units) of measured continuum from calculation [10] (black open circles).

As Fig. 4 illustrates, the results from [18] and [40] both slightly underestimate our measured continuum values. In marked contrast, the trajectory-based calculations from [10] show nearly perfect agreement with the measured data. We hypothesize that this significant improvement stems from the complete and accurate account of induced dipole surface – a feature lacking in the previous studies. Specifically, the calculation by [41] considered only long-range induced dipole contribution, while the calculation by [18] incorporated semi-empirical short-range terms fitted to the far-infrared measurements. The importance of an *ab initio* description of induced dipole is further evidenced by divergence between the trajectory-based and prior theoretical results in the far infrared range, as discussed in [10].

6. Conclusion

The absorption spectra of pure CH_4, CH_4-N_2 and CH_4-CO_2 mixtures have been measured in the 70–354 GHz range using resonator spectrometer. CH_4-CH_4, CH_4-N_2 and CH_4-CO_2 continuum absorption has been determined from the experimental spectra after subtraction of the CH_4 lines and the continuum of pure methane, nitrogen and carbon dioxide. It has been demonstrated that the CH_4-CH_4 and CH_4-N_2 continua in the mm range follow the quadratic frequency dependence. The retrieved continuum coefficients are $5.3(12)\times10^{-16}$ cm/Torr^2 (for pure CH_4) and $3.2(15)\times10^{-15}$ cm/Torr^2 (for CH_4-N_2). However, the quadratic frequency dependence cannot be used for precise modeling of the CH_4-CO_2 continuum coefficient. The uncertainty analysis has revealed that experimental noise (statistical uncertainty) and the line shape parameter uncertainty of methane in HITRAN contribute mostly to the total error.

The small contribution of the line wings to the observed continuum, as well as the agreement of measurement results with the theoretical calculations [7,9,10], confirm that the observed continuum absorption is predominantly of bimolecular nature. To the best of our knowledge, we have made the first room temperature measurements of CH_4, CH_4-N_2 and CH_4-CO_2 continuum in the mm-wave spectral range and thus the first validation of the calculation results in this range.

The work was supported by the Russian State Project No. FFUF-2024-0016. Experimental data were obtained using the research facilities of "CKP-7" (USU N°3589084).

CRediT authorship contribution statement

A.O. Koroleva: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **T.A. Galanina:** Writing – review & editing, Software, Methodology, Formal analysis, Data curation. **I.S. Amerkhanov:** Investigation, Formal analysis, Data curation. **E.A. Serov:** Resources, Methodology, Investigation, Data curation. **A.Yu. Sekacheva:** Validation, Investigation, Data curation. **M. A. Koshelev:** Writing – review & editing, Validation, Methodology, Investigation. **D.N. Chistikov:** Writing – review & editing, Validation, Methodology, Investigation. **A.A. Finenko:** Writing – original draft, Validation, Methodology, Investigation. **M.Yu. Tretyakov:** Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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